

BAAZ & UDAY JKBOSE

2026 CLASS 11TH



Basic Math & Calculus

LECTURE-2

PHYSICS

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Topics To Be Covered

- 1** Revision
- 2** TRIGONOMETRY
- 3** EQUATION
- 4** GRAPH
- 5** IMPORTANT QUESTION

May Allah ≧God≦ grant you
more than your heart hoped for,
open doors you never imagined,
and surprise you with goodness
you never expected

Recap:- Lec...1

Maths

- Maths Bohat Asaan hai

number

+ - × ÷

Fraction

Angle / Power

Trigonometry

Graph / Equation

$\frac{dy}{dx}$ $\int y dx$

Fraction

- $\frac{1}{2} = 0.5$
- $\frac{1}{3} = 0.33$
- $\frac{1}{4} = 0.25$
- $\frac{1}{10} = 0.1$
- $\frac{1}{100} = 0.01$
- $5.42 \times 100 = 542$

Power

- $2^2 = 2 \times 2$
- $x^0 = 1$
- $x^a \cdot x^b = x^{a+b}$
- $(x^a)^b = x^{a \times b}$

$$\frac{1}{0} = \infty$$

$$\frac{1}{\infty} = 0$$

$$\sqrt{2} = 2^{\frac{1}{2}}$$

$$\sqrt{2} = 1.4$$

$$\sqrt{3} = 1.7$$

Angle θ



$$\theta = \frac{\text{Arc}}{\text{Radius}}$$

0 Rad.

$$\rightsquigarrow 180^\circ = \pi \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

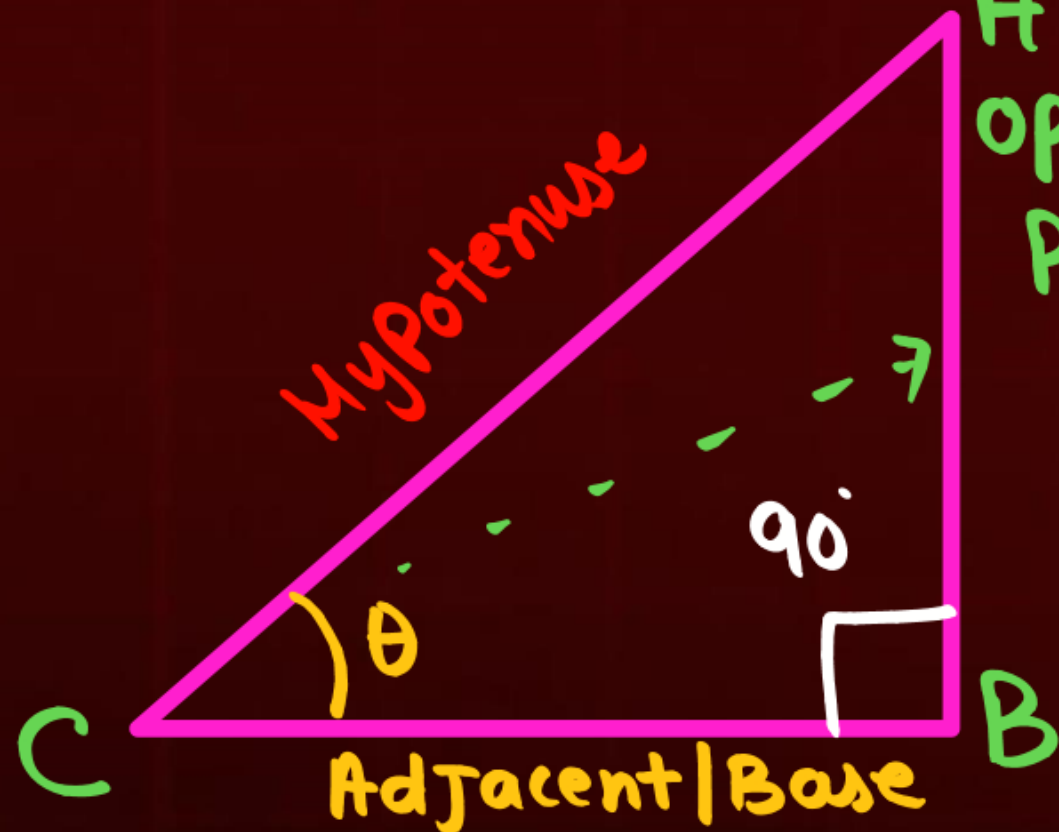
$$1 \text{ rad} = \frac{180^\circ}{\pi}$$

Trigonometry

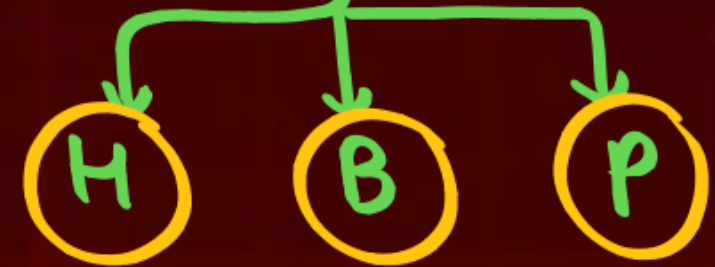
T-1

Lecture no. 2

- Class - 10th chapter - ⑧
- Branch of maths deal with measurement of Right Angle triangle



Pythagoras th



PGT

$$H^2 = B^2 + P^2$$

Measurements

Side

Angle

Sin θ

Cos θ

Tan θ

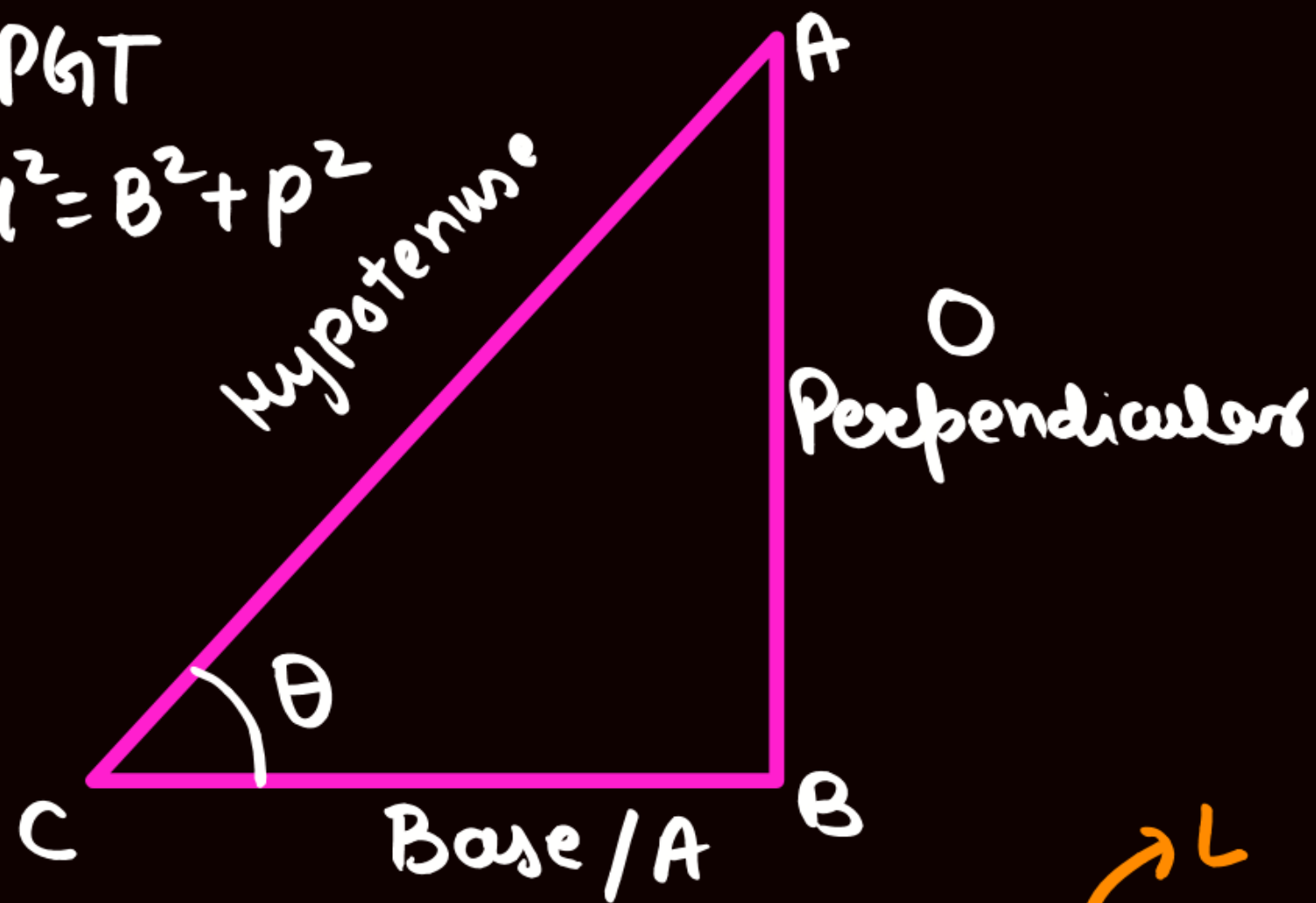
Cosec θ

Sec θ

Cot θ

PGT

$$H^2 = B^2 + P^2$$



$$\# \operatorname{Cosec} \theta = \frac{1}{\sin \theta}$$

$$\# \sec \theta = \frac{1}{\cos \theta}$$

$$\# \cot \theta = \frac{1}{\tan \theta}$$

$$\theta = \frac{\text{Arc}}{\text{R}} \begin{matrix} \nearrow L \\ \searrow L \end{matrix}$$

$$\bullet \sin \theta = \frac{P}{H} = \frac{O}{H} = \frac{AB}{AC}$$

$$\bullet \cos \theta = \frac{B}{H} = \frac{A}{H} = \frac{BC}{AC}$$

$$\bullet \tan \theta = \frac{P}{B} = \frac{O}{A} = \frac{AB}{BC}$$

$$\bullet \operatorname{Cosec} \theta = \frac{H}{P} = \frac{H}{O} = \frac{AC}{AB}$$

$$\bullet \sec \theta = \frac{H}{B} = \frac{H}{A} = \frac{AC}{BC}$$

$$\bullet \cot \theta = \frac{B}{P} = \frac{A}{O} = \frac{BC}{AB}$$

$\sin \theta$

$$\frac{P}{H}$$

$\cos \theta$

$$\frac{B}{H}$$

$\tan \theta$

$$\frac{P}{B}$$

8th

QUESTIONS

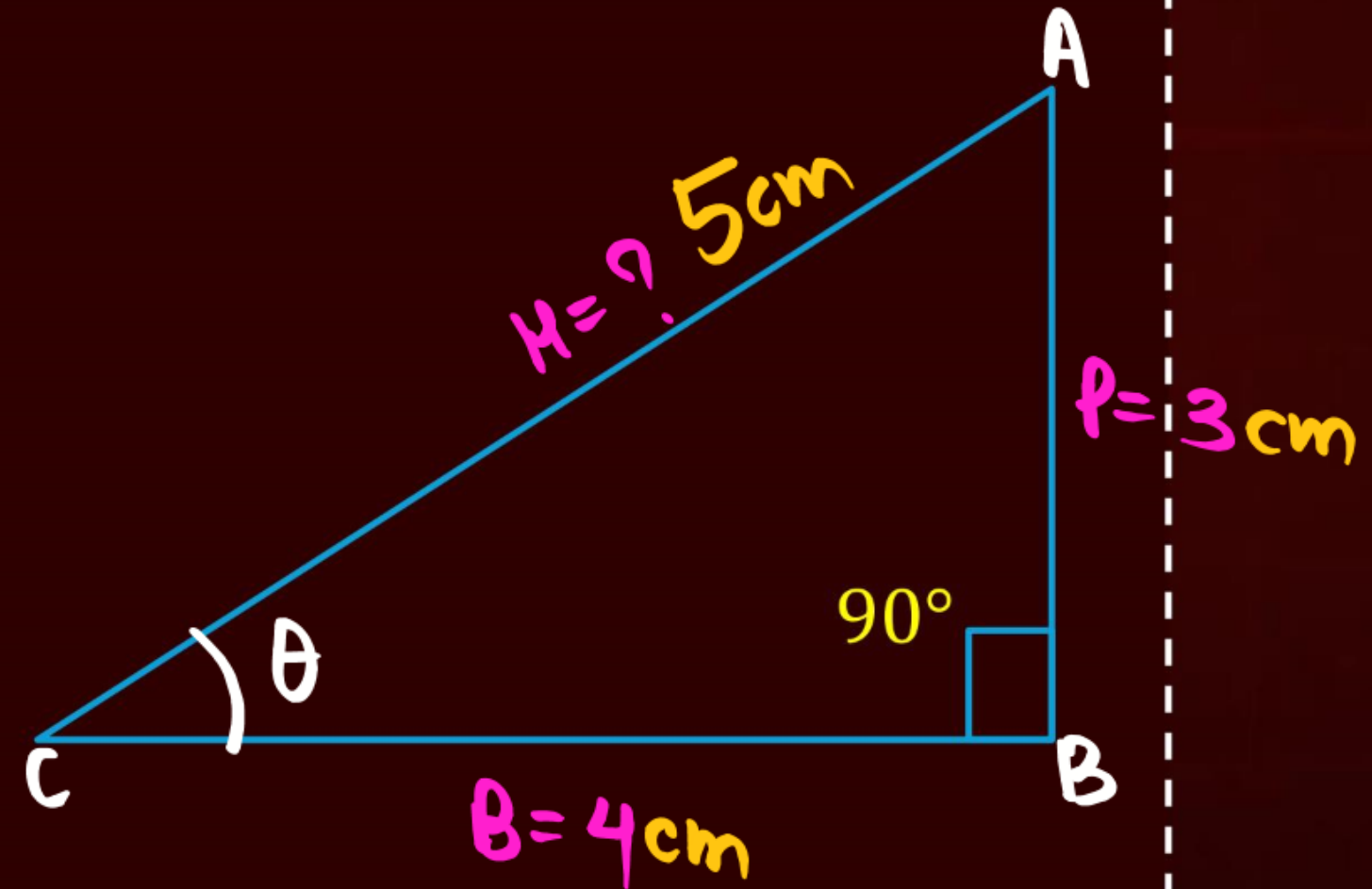
Special triangle (Physics)

H = ?

Sol

$$H^2 = B^2 + P^2$$
$$H^2 = 4^2 + 3^2$$
$$H^2 = 16 + 9$$
$$H^2 = 25$$
$$H^2 = 5^2$$
$$H = \sqrt{5^2} = 5$$
$$\sin \theta = \frac{P}{H} = \frac{3}{5}$$
$$\cos \theta = \frac{B}{H} = \frac{4}{5}$$
$$\tan \theta = \frac{P}{B} = \frac{3}{4}$$
$$\operatorname{cosec} \theta = \frac{H}{P} = \frac{5}{3}$$
$$\sec \theta = \frac{H}{B} = \frac{5}{4}$$
$$\cot \theta = \frac{B}{P} = \frac{4}{3}$$

[Neet]



Is Triangle K Measurement Dimag Main Rakne hai

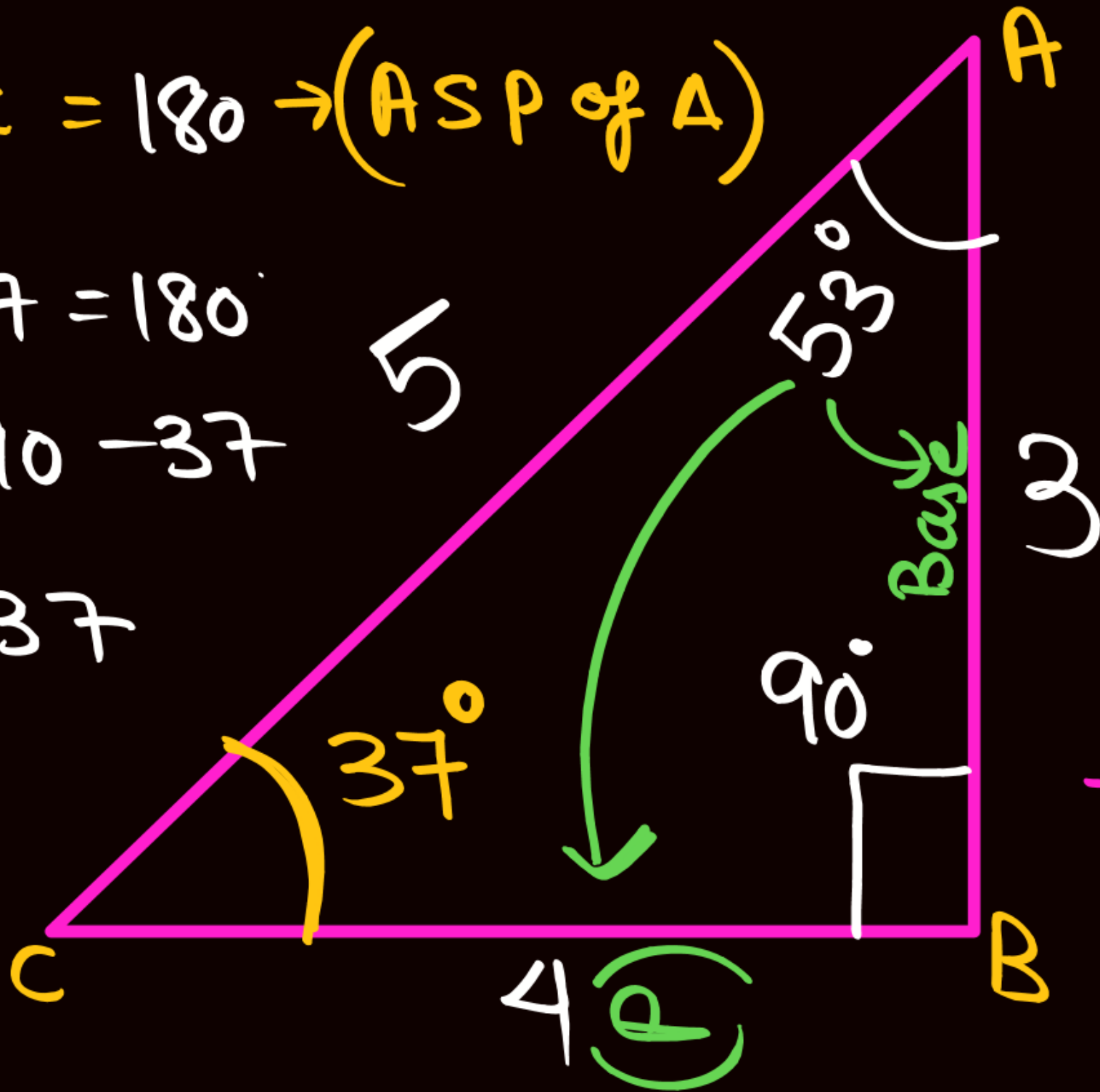
$$\underline{\angle A + \angle B + \angle C = 180} \rightarrow (\text{ASP of } \Delta)$$

$$\angle A + 90^\circ + 37 = 180$$

$$\angle A = 180 - 90 - 37$$

$$\angle A = 90 - 37$$

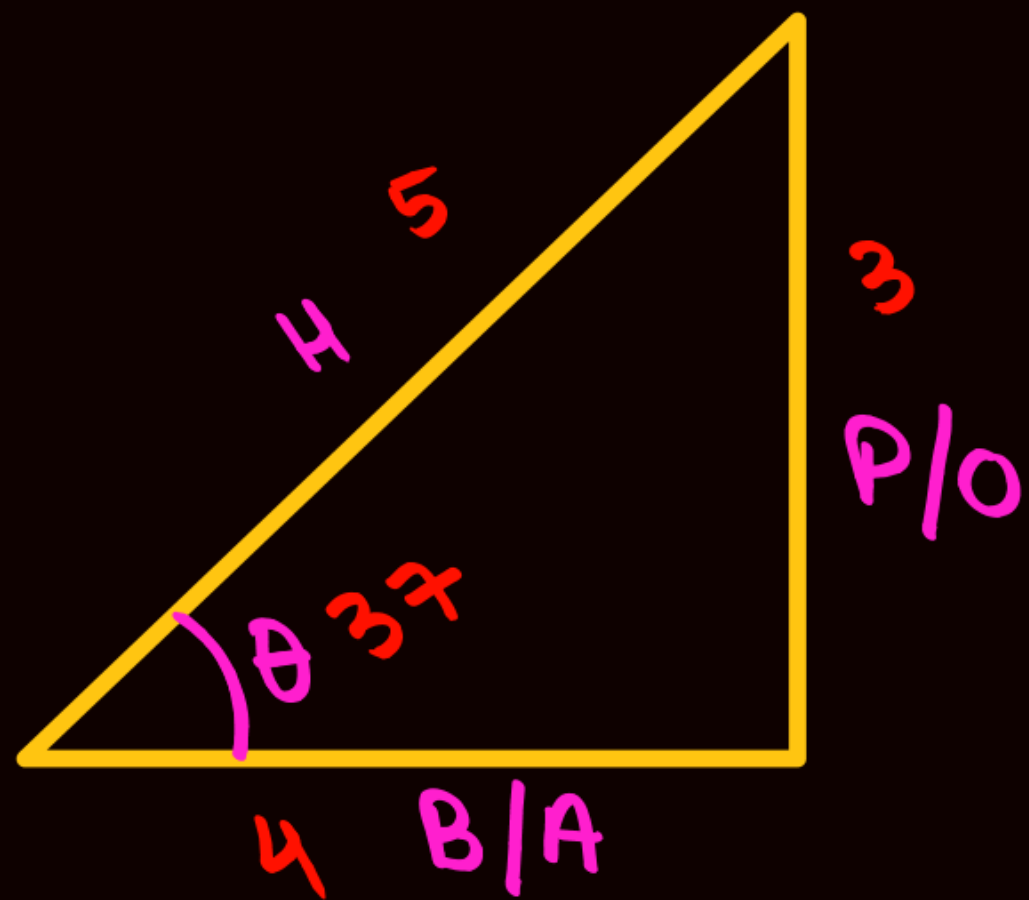
$$\underline{\angle A = 53^\circ}$$



$$\tan 37^\circ = \frac{P}{B} = \frac{3}{4}$$

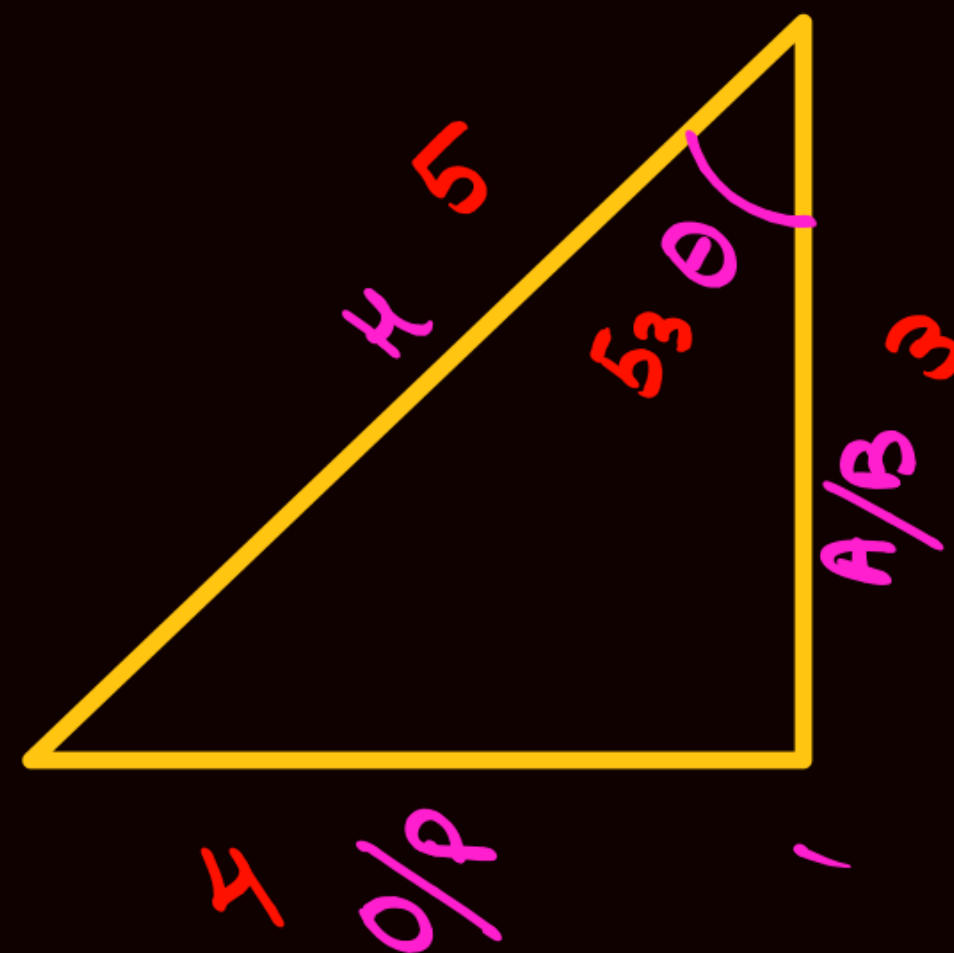
$$\tan 53^\circ = \frac{P}{B} = \frac{4}{3}$$

Note ①



$$\tan 37 = \frac{P}{B} = \frac{3}{4}$$

Note ②



$$\tan 53 = \frac{P}{B} = \frac{4}{3}$$

Trigonometry identity

$$\sin^2 A + \cos^2 A = 1$$

$$\sin^2 \overset{\curvearrowright \theta/B}{\underline{A}} + \cos^2 \overset{\curvearrowright \theta/B}{A} = \textcircled{1}$$

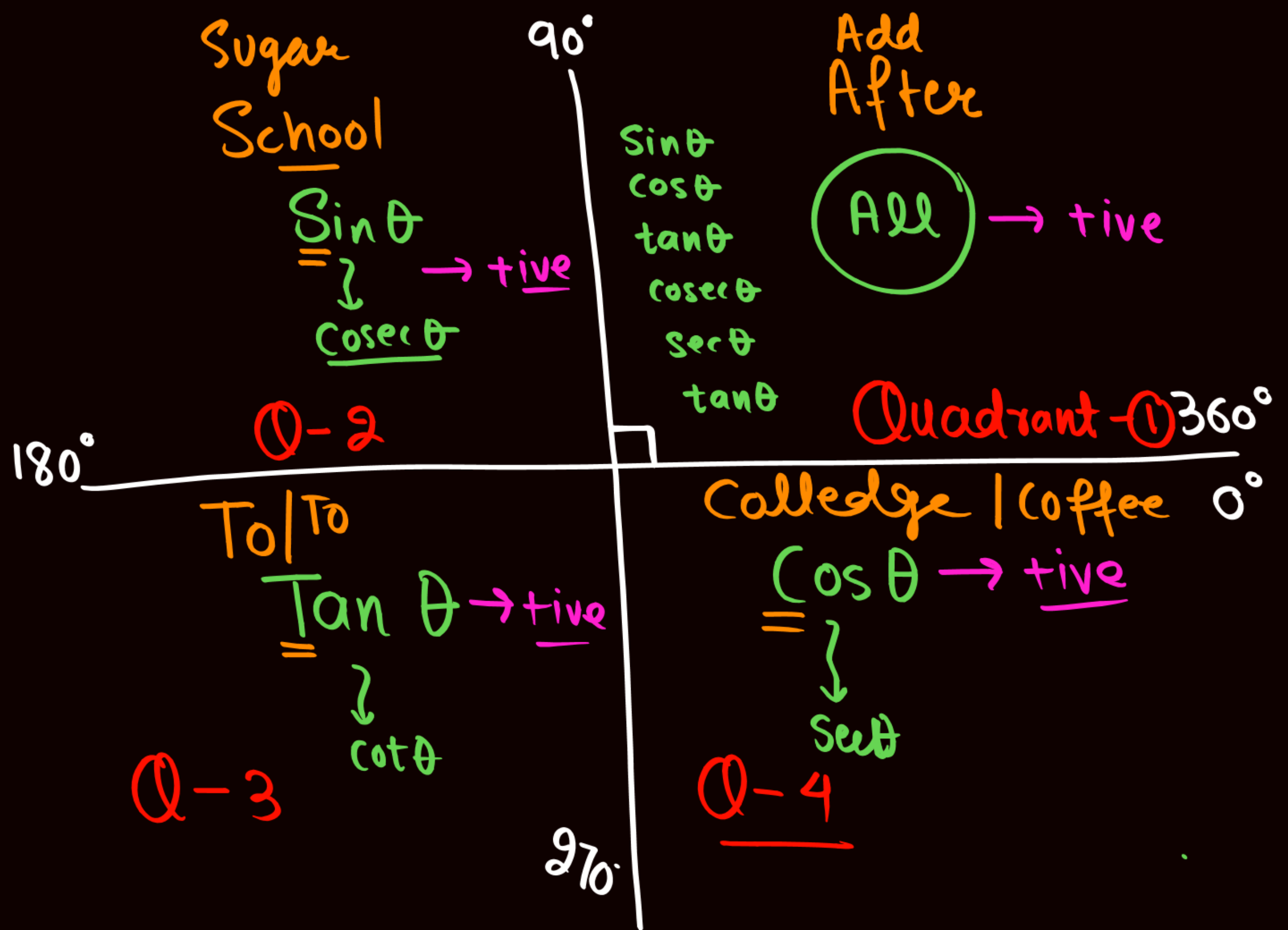
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 B + \cos^2 B = 1$$

Trigonometry Table

$$\begin{array}{ccccc} \frac{0}{4} & \frac{1}{5} & \frac{2}{4} & \frac{3}{5} & \frac{4}{4} \\ 0 & 1 & 2 & 3 & 1 \\ \sqrt{0} & \sqrt{1} & \sqrt{2} & \sqrt{3} & \sqrt{4} \end{array}$$

Angle	0°	30°	45°	60°	90°	120°	135°	150°	180°
sin θ	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
cos θ	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
tan θ	$\frac{0}{1} = 0$	$\frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}$	1	$\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$	$\frac{1}{0} = \infty$	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0



QUESTIONS

$\cos 135^\circ$

- 1 Positive
- 2 Negative ✓
- 3 Zero
- 4 Mujhe nhi aata

Q tan 120° → negative

cos 120° ~~~> negative

Q tan 135° → negative

sin 120° ~~~> positive

Trigonometry



$$\sin 250 \rightarrow 3Q$$

-ive ✓

$$\tan 300 \rightarrow$$

-ive ✓

$$\cos 200 \rightarrow$$

-ive ✓

$$\sec 45 \rightarrow$$

+ive ✓

Q-2

S

Q-1

All

90

360
0

(T)

C ✓

Q-3

Q-4

270

QUESTIONS

Cos 2° is equal

1

1

2

2

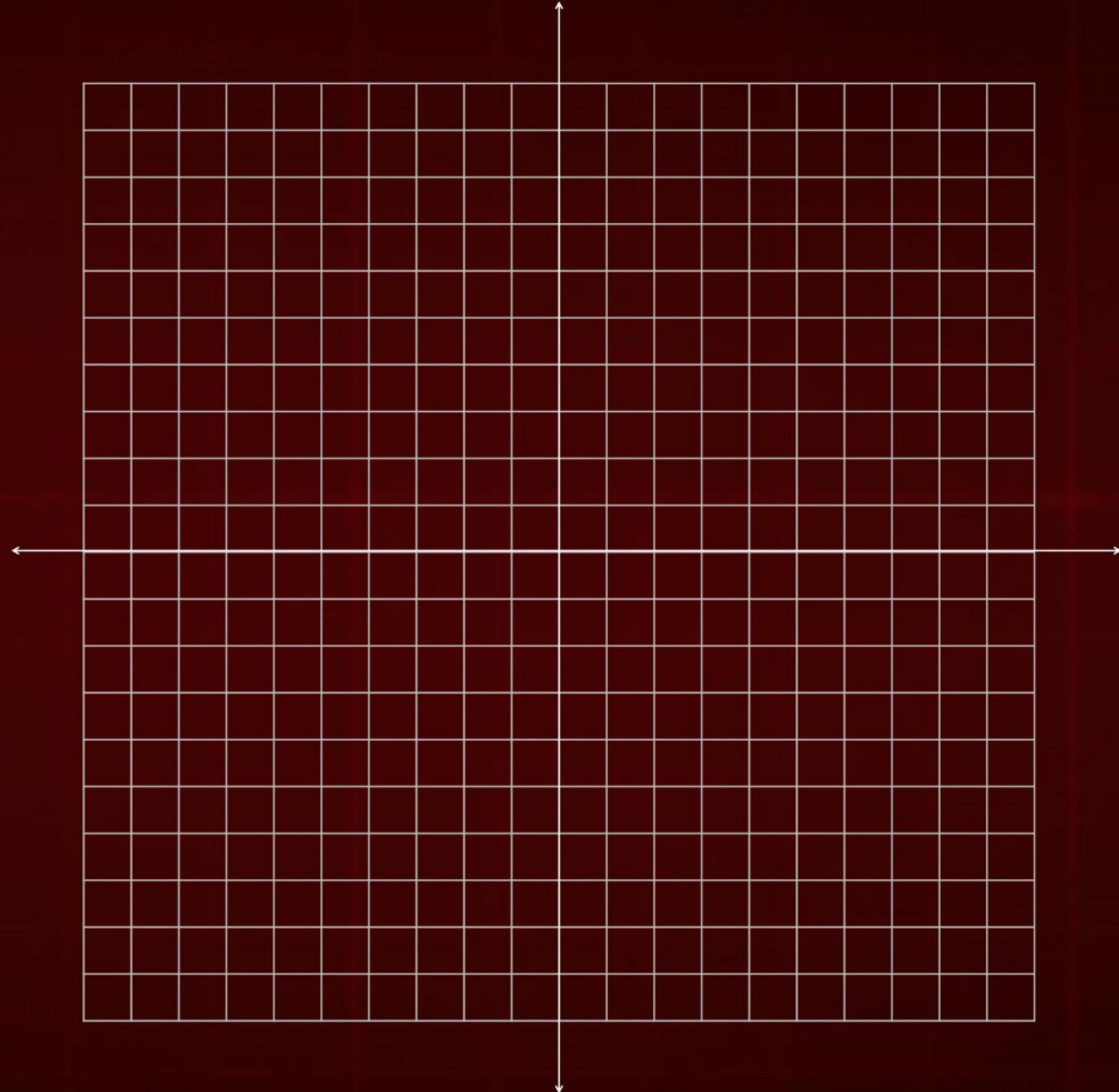
3

3

4

5

Graph



Graph



1. Straight line graph

QUESTIONS

$$y = 3x + 5$$



QUESTIONS

$$y = 3x, \quad C = 0$$



QUESTIONS

$$y = 3x - 2,$$



QUESTIONS

$$y = -3x + 2,$$

QUESTIONS

$$y = -3x - 3,$$



Graph

$$2. y = x^2$$



QUESTIONS

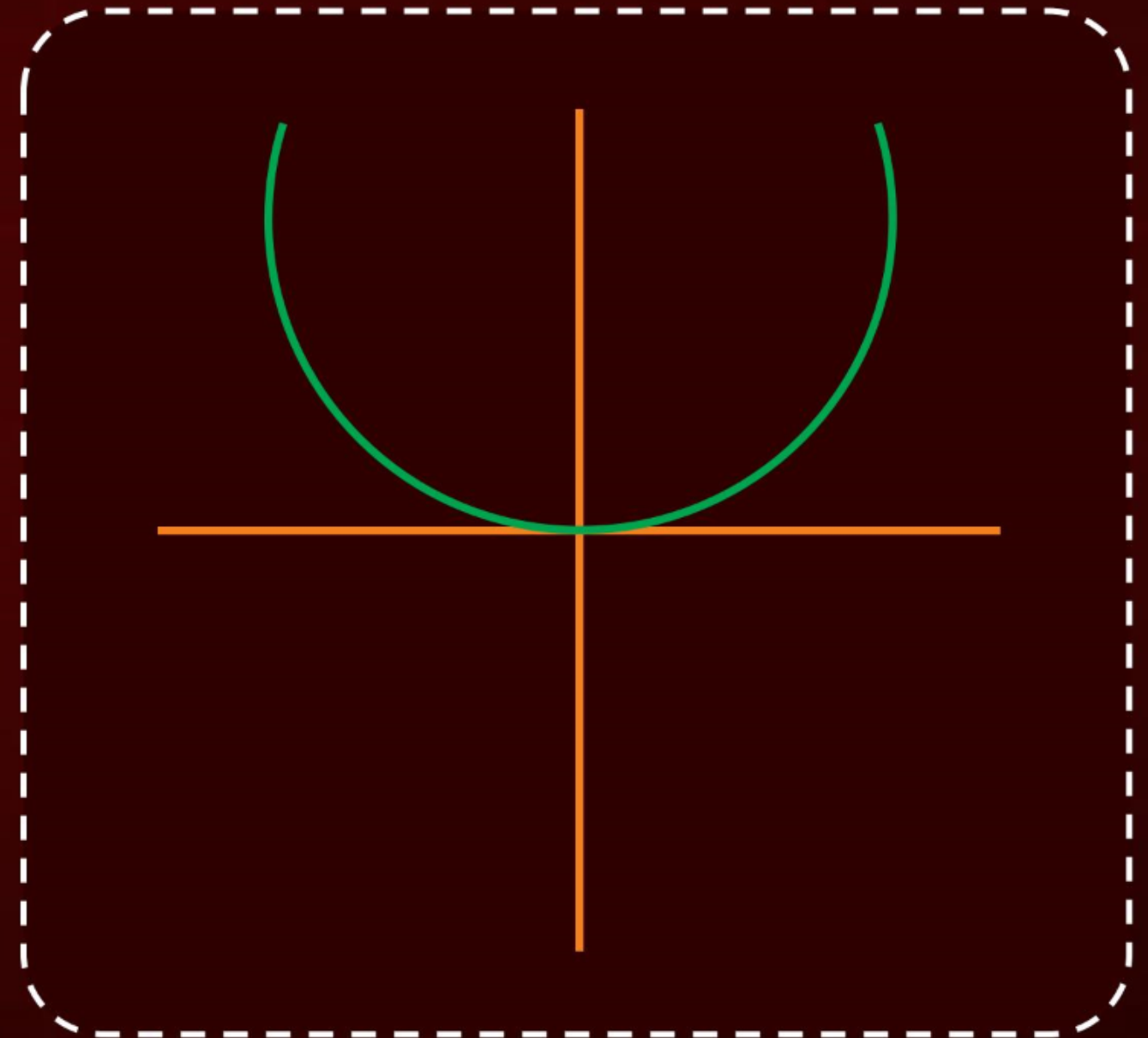
See Graph

1 $y = x^2$

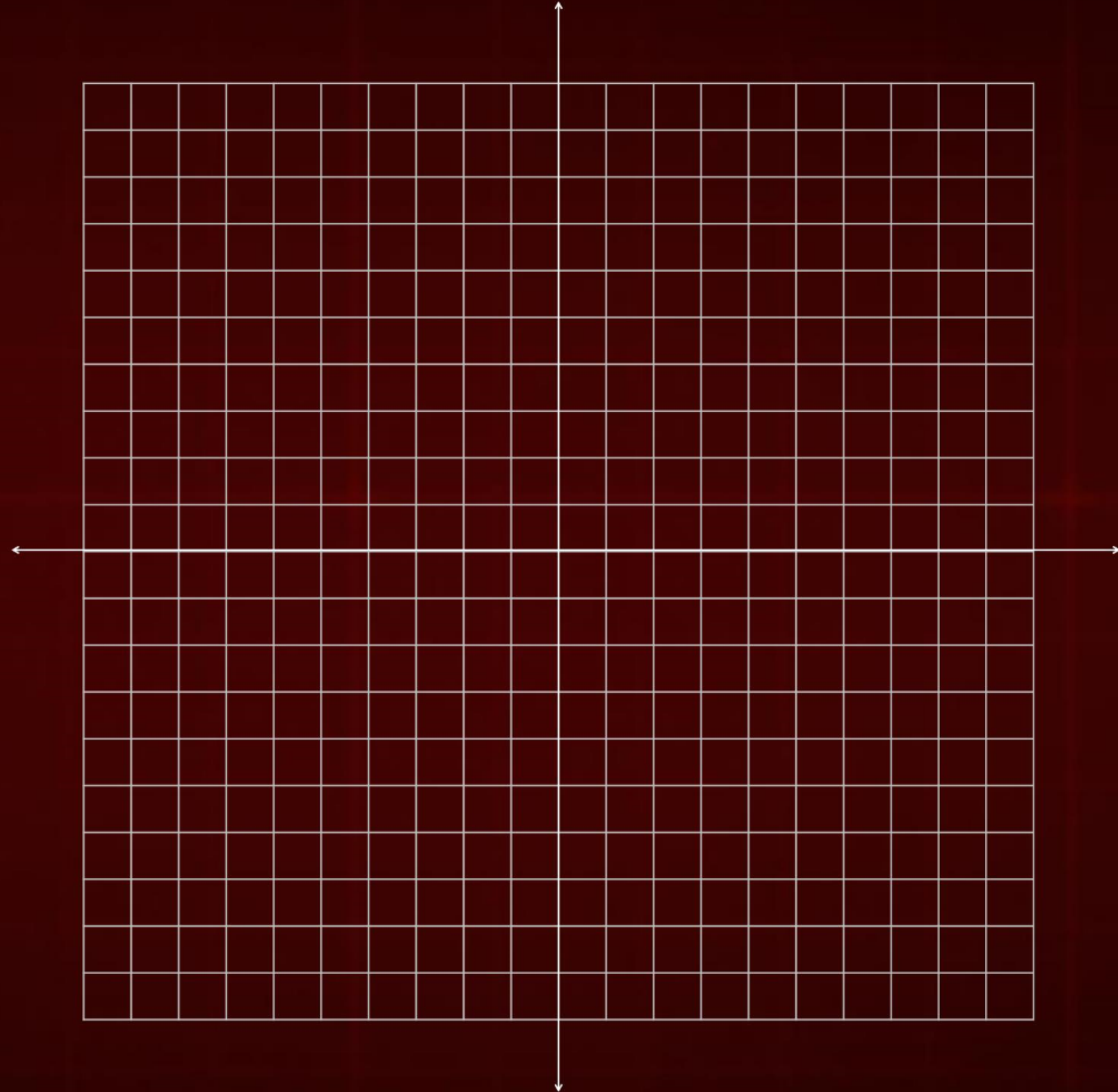
2 $y = mx + c$

3 $y = mx$

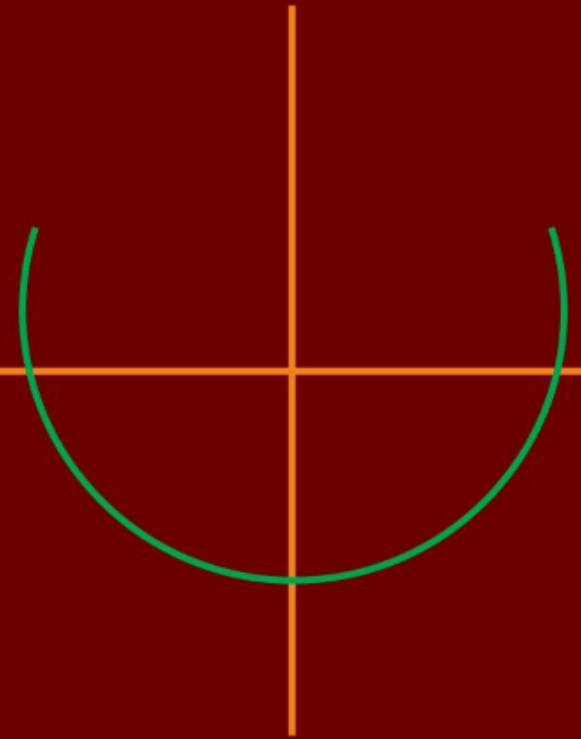
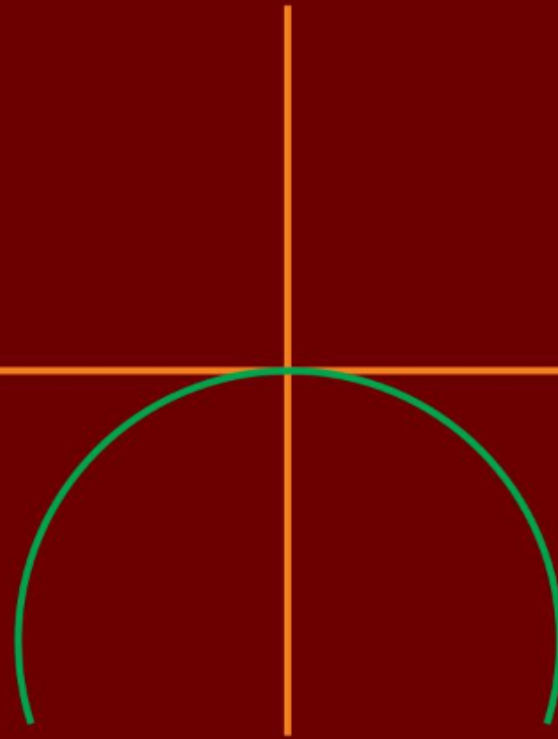
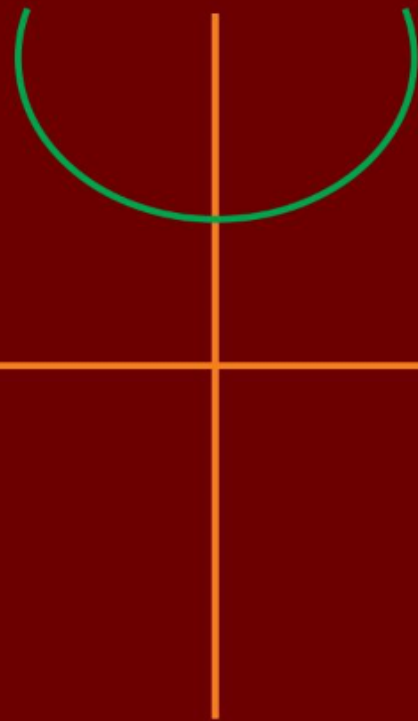
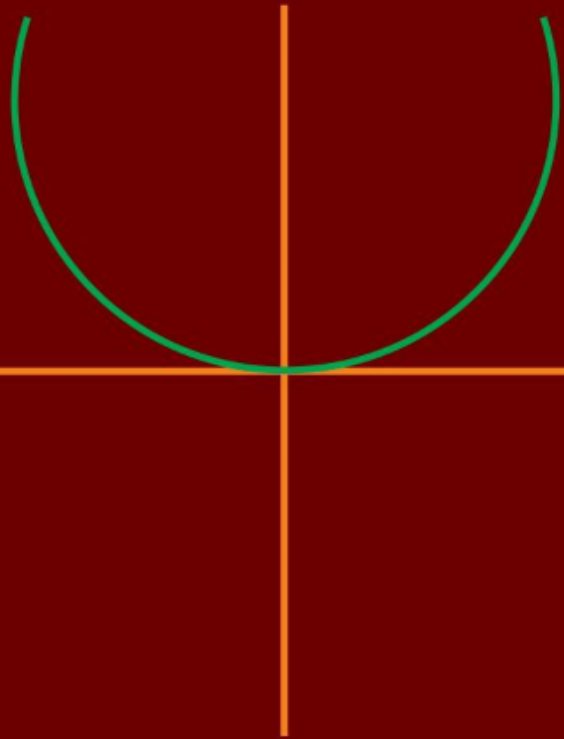
4 $y = \frac{1}{x}$



Graph



QUESTIONS



SUMMARY

- Revision

- Trigonometry

- PGT

- $\sin \theta$

$\cos \theta$

$\tan \theta$

$\operatorname{cosec} \theta$

$\sec \theta$

$\cot \theta$

- $\sin^2 \theta + \cos^2 \theta = 1$

HOMework



$$\underline{\underline{Q}} \quad \underline{\sin} 145^\circ \rightarrow$$

$$\cos 145^\circ \rightarrow$$

$$\tan 200 \rightarrow$$

Thank
You